SHUBHENDU BANERJEE

Data & AI · MSc Big Data & Artificial Intelligence · Halle (Saale), Germany
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PERSONAL DETAILS

Status: MSc student (Big Data & AI), SRH Leipzig — German student visa, eligible for working-student and internship roles
Availability: Open to working-student and internship roles in data & AI
Location: Halle (Saale); open to relocation and hybrid / remote
Languages: English (C1) · Bengali (native) · Hindi (native) · German (A1, actively learning)

PROFILE

Data & AI professional completing an MSc in Big Data & Artificial Intelligence, with 7+ years of prior experience in data operations, quality and coordination. I build and, just as importantly, evaluate AI and data systems: data pipelines, LLM and retrieval (RAG) features, and automation. My focus is less “does it run” and more “can the output be trusted”: knowing when a model works and when a simpler, more robust approach is better. Strong Python and SQL, hands-on with LLMs, embeddings and cloud, and comfortable bridging technical and business perspectives.

KEY SKILLS

AI / ML: LLMs (Claude, ChatGPT, Mistral) | RAG & Embeddings | Vector DBs (Qdrant) | NLP & text classification | Machine Learning | Model evaluation & benchmarking | Prompt engineering
Data & Cloud: Python (pandas, NumPy, scikit-learn, statsmodels) | SQL / DuckDB | Data pipelines & ETL | Data quality & validation | AWS (Lambda, API Gateway, DynamoDB) | Terraform | Docker | Git | Streamlit | Neo4j
Functional: AI/technology evaluation | Requirements & stakeholder communication | Documentation | Process improvement & automation | ISMS / ISO 27001
Learning: Agentic frameworks (LangGraph) | PyTorch | MCP

EDUCATION & CERTIFICATIONS

MSc Computer Science — Big Data & Artificial Intelligence
SRH University, Campus Leipzig, Germany | Oct 2025 – Mar 2027 (expected)
- 1.1 (excellent) in Machine Learning · 1.9 in Advanced Mathematics · passed Agile Project & Risk Management.

Bachelor of Computer Application (BCA)
Dr. A.P.J. Abdul Kalam University, India | 2017 – 2020
- First Division. Programming (C, C++, Java), databases (DBMS/SQL), data structures, mathematics & statistics.

Certification: Professional Certificate in Data Analytics — IIT Kanpur / Simplilearn (SQL, Tableau, Excel, Python) — in progress.

SELECTED PROJECTS

- **AI-Enhanced OSINT — GDELT–KBA Registration Forecasting (research project, team of 4, graded 1.1):** NLP pipeline over about 291K German news records (embeddings, community detection, LLM labelling), tested against official KBA vehicle-registration data with stationarity and Granger-causality tests. An early apparent signal did not survive stationarity correction and was reported transparently as a documented null result. IEEE-format paper. Python · BigQuery · sentence-transformers · statsmodels.
- **Serverless AI Service on AWS (solo):** designed, built and deployed a Python service end to end (API Gateway → Lambda → Secrets Manager → external API → DynamoDB, S3 site) in Terraform, with structured outputs and secure credential handling. github.com/Bshubhendu33/AWS-serverless-news-pipeline
- **H&M Customer Intelligence Pipeline (MSc, team of 5):** pipeline over 1.37M customers / 9.1M transactions: Neo4j graph analytics feeding a classifier (81%→87%, F1 0.87) plus semantic search over a Qdrant vector store. My part: the application/integration layer (Streamlit dashboard). Python · DuckDB · Neo4j · Qdrant · Streamlit · Docker.
- **Customer Segmentation for Marketing (MSc, team of 3):** unsupervised K-Means segmentation of 8,500 credit-card customers into four actionable groups, with EDA, preprocessing, k-selection and a strategy per segment. Python · scikit-learn.
- **Reinforcement Learning — MountainCar (MSc, team of 3):** trained and evaluated a tabular Q-learning agent to ~96% success, with deployment/monitoring thinking. Python · OpenAI Gymnasium · NumPy.

PROFESSIONAL EXPERIENCE

Senior Representative, Operations — Trust & Safety, Data & Quality
Concentrix Daksh Services Pvt. Ltd., Gurgaon, India | Nov 2021 – Aug 2025
- Content/data evaluation against evolving guidelines (YouTube Trust & Safety); built KPI dashboards and reporting used by management for decisions.
- Analysed processes for inefficiencies and led improvements end to end; coordinated across international, cross-functional teams.

Data Analyst / Team Lead (Acting) — AI Training Data
iMerit Technology Services Pvt. Ltd., Ranchi, India | May 2017 – Mar 2021
- Prepared and quality-assured training data for computer-vision, autonomous-driving and NLP models (2D/3D, LiDAR, keypoint annotation); wrote guidelines and adjudicated disagreements to raise inter-annotator agreement.
- Promoted to Acting Team Lead: coordinated the team, ran client communication, and tracked quality and delivery; ISMS / ISO 27001 trained.